True Sigma_1 sentences are provable in Q

The Open Logic Project contributors

openlogic:incompleteness/representability-in-q:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author theorem labelled thm:sigma1-completeness (2100-2104); the proof span 2106-2118 is the author proof environment that immediately follows it.

Editorial note. The exercise directly before the theorem asks the reader to reprove the existential case of lem:delta0-completeness; it belongs to the lemma's delivery, not to this theorem's proof.

Editorial scope. One original result of the Open Logic Project unit incompleteness/representability-in-q, counted once: true Sigma_1 sentences are provable in Q. Only the counted theorem and its own author proof are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

Allowed standard prerequisites (editorial).

- Delta_0-completeness in Q. For every Delta_0 formula B(x) and every n, if N satisfies B(numeral n) then Q proves B(numeral n).. It is the cited engine of the proof (2113-2117); it is the unit's own lemma lem:delta0-completeness.
- Sigma_1 form. A Sigma_1 sentence is a sentence of the form exists x A(x) with A a Delta_0 formula.. It fixes the shape of the sentence the proof analyzes (2106-2110).

Source reference. Original lemma lem-delta0-completeness: sources/olp-incompleteness-representability-in-q/pinned-original-source.tex, source lines 2030–2030. The cited result is separately proved within the same frozen unit and delivered as its own corpus card; the reference is contextual packaging, and the counted proof retains its own argumentative steps in full, so the delivery does not depend on the referenced statement for its completeness.

Theorem 0.1. *If $!A$ is a Σ_1 !!sentence which is true in $\mathfrak{A}N$, then* $\mathrm{Th}Q \vdash !A$.

Proof. If $\exists x !A(x)$ is a Σ_1 !!sentence which is true in $\mathfrak{A}N$, then there exists a natural number n and a variable assignment s such that $s(x) = n$ and $N \models !A(x)[s]$. By standard facts about the satisfaction relation it follows that $N \models !A(\underline{n})$. But $!A(\underline{n})$ is a Δ_0 !!formula, so by lem-delta0-completeness we have that $\mathrm{Th}Q \vdash !A(\underline{n})$, and hence by logic we also have that $\mathrm{Th}Q \vdash \exists[x][!A(x)]$. □

Proof. If $\exists x !A(x)$ is a Σ_1 !!sentence which is true in $\mathfrak{A}N$, then there exists a natural number n and a variable assignment s such that $s(x) = n$ and $N \models !A(x)[s]$. By standard facts about the satisfaction relation it follows that $N \models !A(\underline{n})$. But $!A(\underline{n})$ is a Δ_0 !!formula, so by lem-delta0-completeness we have that $\mathrm{Th}Q \vdash !A(\underline{n})$, and hence by logic we also have that $\mathrm{Th}Q \vdash \exists[x][!A(x)]$. $\square$